

Q-Learning with Batch Sampling: A Convergence Analysis

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Abstract

This report studies a batched variant of tabular Q-learning in which several transition samples are aggregated before an update is applied. The motivation is that single-sample Q-learning is highly stochastic, whereas averaging over a batch can reduce update noise and may accelerate convergence.

On the theoretical side, the analysis is formulated as a stochastic approximation problem for an empirical Bellman operator that updates a batch of state-action coordinates in each iteration. The expectation of this operator is shown to be a convex combination of the Bellman optimality operator and the identity. Based on a Euclidean acute-angle argument, a pseudogradient condition is established, and a second-moment bound is derived. Under standard boundedness and step-size assumptions, together with the condition $\sqrt{d}\gamma < 1$, almost sure convergence to the optimal action-value function q^* follows. The resulting drift constant depends linearly on the batch size m , which explains why larger batches should improve the convergence rate.

On the empirical side, this theoretical prediction is examined on the tabular FrozenLake environment following an experimental setting. A batch-size comparison shows that larger values of m lead to a faster reduction of the distance to Q^* , especially in the initial phase of learning.

Taken together, the theoretical and empirical results support the central claim of this report: in the tabular setting considered here, batch sampling improves the convergence behavior of Q-learning.

1 Introduction

Q-learning is one of the fundamental algorithms in reinforcement learning. In its classical tabular form, it updates the value of a single state-action pair from a single observed transition [10]. This update rule is simple and effective, but it is also noisy because every iteration depends on one stochastic sample and can therefore fluctuate strongly, especially in environments with random transitions [10].

A natural way to reduce this noise is to replace the single-sample target by an average over several sampled transitions. This idea is closely related to experience replay, where past transitions are stored in a replay buffer and sampled again during learning [5, 6]. In practical reinforcement learning, replay and mini-batch updates are standard tools [6]. Nevertheless, the precise way in which batching affects convergence is less transparent than in the classical single-sample setting [11].

The present report investigates this question for tabular Q-learning. The guiding hypothesis is that averaging several sampled Bellman targets strengthens the systematic movement toward the optimal fixed point and therefore leads to faster convergence. On the theoretical side, this hypothesis is studied through a batch-sampled empirical Bellman operator within a stochastic-approximation framework [12]. On the empirical side, the resulting prediction is tested in a batch-update setting on FrozenLake, where the influence of the batch size can be examined directly.

The main contribution of the theoretical part is a convergence proof for a batched tabular Q-learning scheme. The proof shows that the expected update direction is a pseudogradient

direction for a quadratic Lyapunov function and that the strength of the resulting drift grows linearly with the batch size. This leads to almost sure convergence under standard assumptions and makes the role of the batch parameter explicit in the convergence rate.

The main contribution of the empirical part is a study of the batch-size effect. A controlled batch experiment isolates the parameter m and shows that larger batches reduce the distance to the optimal action-value function more rapidly, especially in the early phase of learning. A normalized comparison across discount factors further shows that the advantage of the best-performing batched configuration remains visible across a broader range of discount factors. All experimental curves reported in the main text are aggregated across multiple random seeds. In the batch-size comparison, shaded regions indicate one standard deviation, while in the cross-gamma comparison only the mean curves are shown to keep the figure readable.

The report is organized as follows. Section 3 develops the theoretical framework and presents the convergence analysis. Section 4 reports the empirical evaluation and relates the observed behavior to the theoretical predictions. The final sections discuss the main implications and summarize the results.

2 Related Work

Q-Learning was introduced by Watkins [13] and later analyzed in detail by Watkins and Dayan [14], who proved almost sure convergence in the tabular setting under repeated sampling and standard step-size conditions. Their paper already notes extensions in which many Q -values may be changed in one iteration rather than only one, which makes it a natural starting point for studying batched updates [14]. The broader reinforcement learning background used in this work, including finite Markov decision processes and Bellman optimality equations, is standardly covered by Sutton and Barto [10].

A second line of foundational work studies Q-learning through stochastic approximation. Tsitsiklis [12] developed an asynchronous stochastic-approximation analysis of Q-learning and established convergence under assumptions more general than those used in the original proof of Watkins and Dayan. Closely related dynamic-programming and contraction-based perspectives are presented in the work of Bertsekas and Tsitsiklis [1]. Since the present report also formulates the update as a stochastic-approximation scheme and studies its drift through a Lyapunov argument, these works provide the main theoretical context.

The practical motivation for batch sampling comes from experience replay. Lin [5] introduced experience replay as a mechanism for storing and reusing past experience during reinforcement learning. The idea later became a standard ingredient of deep reinforcement learning through the Deep Q-Network architecture of Mnih et al. [6], where replay buffers and mini-batch updates were used to stabilize and improve learning. These works do not provide the convergence analysis pursued here, but they explain why replay and batched updates are of central practical interest.

More recent work has begun to analyze replay-based sampling itself. Szlak and Shamir [11] provide convergence-rate guarantees for tabular Q-learning with experience replay and show theoretically and empirically when replay may improve performance. Furthermore, the role of replay has also been studied systematically in deep Q-learning settings by Fedus et al. [2], where replay capacity and replay ratio were shown to have a substantial influence on performance. These papers are closely related to the present report, since they study how replayed data can alter the speed of convergence.

Beyond uniform replay, several papers investigate how replay batches should be chosen. Schaul et al. [9] proposed prioritized experience replay, in which transitions with larger learning signal are replayed more frequently, and reported improved learning efficiency in

deep Q-networks. Lahire et al. [4] later cast replay sampling as an importance-sampling problem, derived an optimal sampling distribution, and proposed Large Batch Experience Replay (LaBER), which uses large candidate batches together with focused updates. These papers are primarily concerned with deep reinforcement learning, but they show that the design of the replay batch can have a strong influence on learning speed and stability.

The present report is positioned between these lines of work. On the one hand, it follows the classical stochastic-approximation tradition by deriving a convergence result for a tabular batch-update scheme. On the other hand, it is motivated by the practical replay literature and investigates whether averaging multiple sampled transitions leads to faster convergence in practice. In this sense, the report connects foundational Q-learning theory with the modern use of replay-based mini-batch updates.

3 Theoretical Foundations and Convergence Analysis

3.1 Markov Decision Processes

We consider a finite Markov decision process with state space $\mathcal{S}$, action space $\mathcal{A}$, transition dynamics, reward function, and discount factor $\gamma \in [0, 1)$. For a state-action pair $(s, a) \in \mathcal{S} \times \mathcal{A}$, the action-value function $Q(s, a)$ denotes the expected discounted return obtained by taking action a in state s and acting optimally thereafter [10].

Throughout the report, the tabular action-value function is identified with a vector $q \in \mathbb{R}^d$, where

$$d = |\mathcal{S}| |\mathcal{A}| \quad (1)$$

is the number of state-action pairs. No distinction will be made between the tabular representation $Q(s, a)$ and its vector form q .

The central object in the analysis is the Bellman optimality operator. For $q \in \mathbb{R}^d$, it is defined componentwise by

$$(B[q])(s, a) = \mathbb{E} \left[R(s, a, S') + \gamma(1 - D(s, a, S')) \max_{a' \in \mathcal{A}} q(S', a') \mid s, a \right], \quad (2)$$

where S' is the next state, $R(s, a, S')$ is the immediate reward, and $D(s, a, S') \in \{0, 1\}$ is a terminal-state indicator [10]. The optimal action-value function q^* is characterized by the fixed-point equation [1, 10]

$$q^* = B[q^*]. \quad (3)$$

Classical one-step Q-learning approximates this fixed point from individual sampled transitions [10, 14]. Its convergence can also be understood within the broader framework of stochastic iterative dynamic programming algorithms [3]. If in iteration n a single transition $(S_n, A_n, R_n, S'_n, D_n)$ is observed, then the update reads

$$Q_{n+1}(S_n, A_n) = Q_n(S_n, A_n) + \alpha_n(S_n, A_n) \left(R_n + \gamma(1 - D_n) \max_{a'} Q_n(S'_n, a') - Q_n(S_n, A_n) \right), \quad (4)$$

while all other components remain unchanged. The step size $\alpha_n(S_n, A_n)$ determines how strongly the new sample affects the current estimate.

The method studied here replaces this single target by an averaged target over multiple sampled transitions. In that sense, batch sampling can be viewed as a tabular abstraction of replay-buffer learning, where several transitions are drawn, their Bellman targets are aggregated, and the update is based on this averaged information rather than on a single sample [5, 6, 11].

3.2 Empirical Bellman Operator with Batch Sampling

The theoretical analysis is based on an empirical Bellman operator that averages over a finite batch of transition samples. Let

$$E_n = \{\xi_{n,j}\}_{j=1}^{M_n}, \quad \xi_{n,j} = (s_{n,j}, a_{n,j}, r_{n,j}, s'_{n,j}, d_{n,j}), \quad (5)$$

be the set of samples available in iteration n . For each state-action pair (s, a) , define

$$C_n(s, a) = \{j \in \{1, \dots, M_n\} : (s_{n,j}, a_{n,j}) = (s, a)\} \quad (6)$$

as the index set of all batch elements associated with that pair.

The empirical Bellman operator T_{E_n, C_n} is defined by

$$(T_{E_n, C_n}[q])(s, a) = \begin{cases} \frac{1}{|C_n(s, a)|} \sum_{j \in C_n(s, a)} \left(r_{n,j} + \gamma(1 - d_{n,j}) \max_{a' \in \mathcal{A}} q(s'_{n,j}, a') \right), & |C_n(s, a)| > 0, \\ q(s, a), & |C_n(s, a)| = 0. \end{cases} \quad (7)$$

This operator has two important features. First, only those coordinates are updated for which the batch contains samples. Second, whenever several samples belong to the same state-action pair, their Bellman targets are averaged before the update is applied. The classical one-step update appears as the special case in which exactly one sample is available for exactly one coordinate.

The operator in (7) serves as a mathematical abstraction of batched replay updates [11]. In a practical replay-buffer implementation, batches are sampled from past experience and therefore depend on the trajectory distribution [5, 6]. In the theoretical analysis, this is replaced by a controlled sampling model that isolates the effect of the batch size m .

3.3 Assumptions

The convergence analysis is based on the following assumptions.

Assumption 1. The state and action spaces are finite, and the rewards are uniformly bounded. That is, there exists a constant $\bar{R} > 0$ such that

$$|R(s, a, S')| \leq \bar{R} \quad \text{for all } (s, a, S'). \quad (8)$$

Assumption 2. In each iteration, conditioned on the past filtration $\mathcal{F}_{n-1}$, exactly m different coordinates are selected uniformly without replacement from the d state-action pairs. Hence, for every (s, a) ,

$$\mathbb{P}(|C_n(s, a)| > 0 \mid \mathcal{F}_{n-1}) = \frac{m}{d}. \quad (9)$$

Assumption 3. Conditioned on $\mathcal{F}_{n-1}$ and on $|C_n(s, a)| > 0$, the averaged empirical target for (s, a) is unbiased for the current iterate. That is, for every $\mathcal{F}_{n-1}$ -measurable $q \in \mathbb{R}^d$,

$$\mathbb{E}[(T_{E_n, C_n}[q])(s, a) \mid \mathcal{F}_{n-1}, |C_n(s, a)| > 0] = (B[q])(s, a). \quad (10)$$

Assumption 4. The step sizes $(\alpha_n)_{n \geq 1}$ satisfy the Robbins–Monro conditions [8]

$$\alpha_n > 0, \quad \sum_{n=1}^{\infty} \alpha_n = \infty, \quad \sum_{n=1}^{\infty} \alpha_n^2 < \infty. \quad (11)$$

Assumptions 1 and 4 describe bounded rewards and the Robbins–Monro conditions on step sizes, and are standard in the convergence literature on tabular Q-learning and stochastic approximation [8, 10]. By contrast, Assumptions 2 and 3 are not standard in exactly this form, but are introduced here to model the specific batch-update scheme studied in this report. In particular, the uniform selection of m coordinates and the corresponding unbiasedness condition are assumptions that isolate the effect of the batch size m in an analytically tractable way. The parameter m measures how many state-action coordinates are updated in a single iteration. Increasing m therefore increases the amount of averaged Bellman information entering the update.

3.4 Expectation of the Empirical Bellman Operator

The first step is to relate the empirical operator (7) to the exact Bellman operator (2).

Theorem 1. Under Assumptions 1 to 3, the identity

$$\mathbb{E}[T_{E_n, C_n}[q] \mid \mathcal{F}_{n-1}] = \frac{m}{d} B[q] + \left(1 - \frac{m}{d}\right) q \quad (12)$$

holds for every $\mathcal{F}_{n-1}$ -measurable $q \in \mathbb{R}^d$.

Proof. Fix an arbitrary $(s, a) \in \mathcal{S} \times \mathcal{A}$. By definition,

$$(T_{E_n, C_n}[q])(s, a) = \mathbf{1}_{\{|C_n(s, a)| > 0\}} Y_n(s, a) + \mathbf{1}_{\{|C_n(s, a)| = 0\}} q(s, a), \quad (13)$$

where $Y_n(s, a)$ denotes the averaged empirical target for the coordinate (s, a) . Taking conditional expectations with respect to $\mathcal{F}_{n-1}$ and using the conditional law of total expectation gives

$$\mathbb{E}[(T_{E_n, C_n}[q])(s, a) \mid \mathcal{F}_{n-1}] = \mathbb{P}(|C_n(s, a)| > 0 \mid \mathcal{F}_{n-1}) \mathbb{E}[Y_n(s, a) \mid \mathcal{F}_{n-1}, |C_n(s, a)| > 0] + \mathbb{P}(|C_n(s, a)| = 0 \mid \mathcal{F}_{n-1}) q(s, a) \quad (14)$$

Assumptions 2 and 3 imply

$$\mathbb{E}[(T_{E_n, C_n}[q])(s, a) \mid \mathcal{F}_{n-1}] = \frac{m}{d} (B[q])(s, a) + \left(1 - \frac{m}{d}\right) q(s, a). \quad (15)$$

Since this is valid for every coordinate, the vector identity (12) follows. $\square$

Equation (12) is the central structural relation of the analysis. It shows that the expected empirical operator is a convex combination of the exact Bellman operator and the identity. The Bellman part is weighted by m/d , so the influence of the fixed-point dynamics grows linearly with the batch size.

3.5 Formulation as a Stochastic Approximation Scheme

The batch update can now be written as a stochastic approximation iteration [8, 12]:

$$q_n = q_{n-1} + \alpha_n (T_{E_{n-1}, C_{n-1}}[q_{n-1}] - q_{n-1}). \quad (16)$$

Equivalently, define the search direction

$$s_n = q_{n-1} - T_{E_{n-1}, C_{n-1}}[q_{n-1}], \quad (17)$$

so that

$$q_n = q_{n-1} - \alpha_n s_n. \quad (18)$$

To analyze convergence, consider the quadratic error function

$$J(q) = \frac{1}{2} \|q - q^*\|_2^2. \quad (19)$$

$$\nabla J(q) = q - q^*. \quad (20)$$

The gradient of J is Lipschitz continuous with constant $L = 1$, since

$$\|\nabla J(q) - \nabla J(\tilde{q})\|_2 = \|q - \tilde{q}\|_2 \quad \text{for all } q, \tilde{q} \in \mathbb{R}^d.$$

This simple smoothness property is used implicitly in the descent estimate for J that leads to the Lyapunov recursion below. Thus, the convergence proof reduces to two tasks: showing that the mean search direction points toward q^* , and bounding its second moment [7, 12].

3.6 Acute-Angle Argument

The geometric core of the proof is an acute-angle argument in Euclidean space, stated here in the finite-dimensional form used in this report and motivated by the pseudogradient perspective in stochastic approximation [7].

Theorem 2. Let $x_1, x_2, x_* \in \mathbb{R}^d$ and $\lambda \in [0, 1)$ satisfy

$$\|x_2 - x_*\|_2 \leq \lambda \|x_1 - x_*\|_2. \quad (21)$$

Then

$$\langle x_* - x_1, x_2 - x_1 \rangle \geq (1 - \lambda) \|x_1 - x_*\|_2^2. \quad (22)$$

Proof. We decompose

$$x_2 - x_1 = (x_* - x_1) - (x_* - x_2). \quad (23)$$

Therefore,

$$\langle x_* - x_1, x_2 - x_1 \rangle = \|x_1 - x_*\|_2^2 - \langle x_* - x_1, x_* - x_2 \rangle. \quad (24)$$

By the Cauchy–Schwarz inequality and (21),

$$\langle x_* - x_1, x_* - x_2 \rangle \leq \|x_1 - x_*\|_2 \|x_2 - x_*\|_2 \leq \lambda \|x_1 - x_*\|_2^2. \quad (25)$$

Substituting this bound into (24) proves (22). $\square$

Theorem 2 formalizes the geometric idea that if x_2 is sufficiently closer to x_* than x_1 in the Euclidean norm, then the direction $x_2 - x_1$ forms an acute angle with the error direction $x_* - x_1$.

Figure 1 illustrates the content of Theorem 2 geometrically. The Euclidean ball around x_* with radius $\|x_1 - x_*\|_2$ describes the set of points that are closer to the limit x_* than the current iterate x_1 . Any point x_2 inside this ball satisfies the acute-angle condition, so the direction $x_2 - x_1$ points toward x_* in a pseudogradient sense. The figure also shows why this reasoning is specific to the Euclidean norm. In the maximum norm, points may be closer to x_* without lying in the correct half-space. This is why the proof uses the Bellman contraction in $\|\cdot\|_\infty$ together with norm equivalence to derive a sufficient condition in $\|\cdot\|_2$.

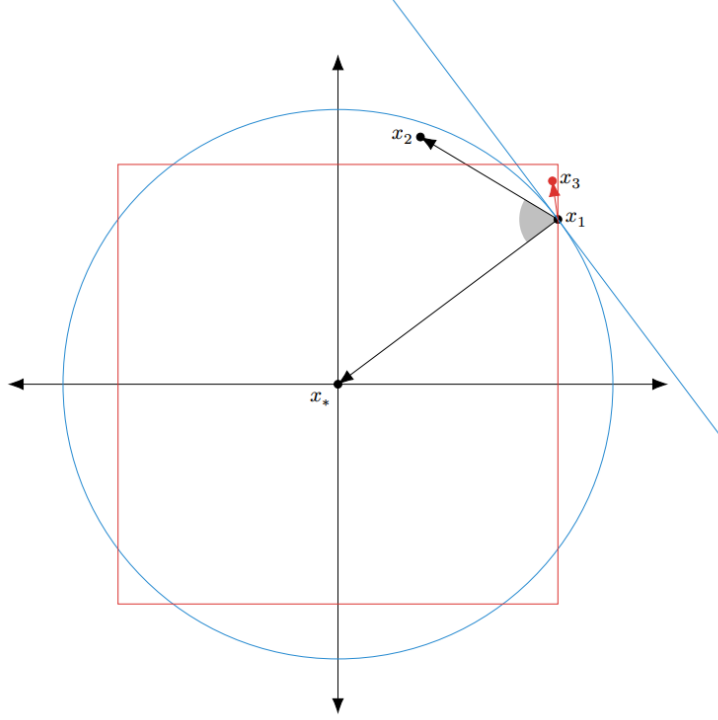


Figure 1: Geometric interpretation of the acute-angle argument. The blue circle contains all points that are closer to x_* than x_1 in the Euclidean norm. Any such point, for example x_2 , lies in the half-space that forms an acute angle with the error direction $x_* - x_1$. By contrast, the red square illustrates that an analogous statement does not hold for the maximum norm: a point such as x_3 may be closer to x_* in $\|\cdot\|_\infty$ while still failing the acute-angle condition.

The Bellman optimality operator is a γ -contraction in the supremum norm [1, 10]:

$$\|B[q] - B[\tilde{q}]\|_\infty \leq \gamma \|q - \tilde{q}\|_\infty \quad \text{for all } q, \tilde{q} \in \mathbb{R}^d. \quad (26)$$

Choosing $\tilde{q} = q^*$ and using (3) yields

$$\|B[q] - q^*\|_\infty \leq \gamma \|q - q^*\|_\infty. \quad (27)$$

Since all norms are equivalent on $\mathbb{R}^d$,

$$\|x\|_\infty \leq \|x\|_2 \leq \sqrt{d} \|x\|_\infty \quad \text{for all } x \in \mathbb{R}^d. \quad (28)$$

Consequently,

$$\|B[q] - q^*\|_2 \leq \sqrt{d} \gamma \|q - q^*\|_2. \quad (29)$$

Hence, $B[q]$ is closer to q^* than q in the Euclidean norm whenever

$$\sqrt{d} \gamma < 1 \quad (30)$$

Applying Theorem 2 with

$$x_1 = q, \quad x_2 = B[q], \quad x_* = q^*, \quad \lambda = \sqrt{d} \gamma \quad (31)$$

gives the key estimate

$$\langle q - q^*, q - B[q] \rangle \geq (1 - \sqrt{d} \gamma) \|q - q^*\|_2^2. \quad (32)$$

3.7 Pseudogradient Condition

We now use the expectation identity and the acute-angle estimate to show that the mean search direction is a pseudogradient direction for J [7].

Theorem 3. Under Assumptions 1 to 4 and condition (30),

$$\langle \nabla J(q), \mathbb{E}[q - T_{E_n, C_n}[q] \mid \mathcal{F}_{n-1}] \rangle \geq \frac{m}{d} (1 - \sqrt{d} \gamma) \|q - q^*\|_2^2 \quad (33)$$

for all $\mathcal{F}_{n-1}$ -measurable $q \in \mathbb{R}^d$.

Proof. By (20) and (12),

$$\langle \nabla J(q), \mathbb{E}[q - T_{E_n, C_n}[q] \mid \mathcal{F}_{n-1}] \rangle = \left\langle q - q^*, q - \frac{m}{d} B[q] - \left(1 - \frac{m}{d}\right) q \right\rangle. \quad (34)$$

This simplifies to

$$\langle \nabla J(q), \mathbb{E}[q - T_{E_n, C_n}[q] \mid \mathcal{F}_{n-1}] \rangle = \frac{m}{d} \langle q - q^*, q - B[q] \rangle. \quad (35)$$

The claim follows immediately from (32). $\square$

The factor m/d in (33) is decisive. It shows that larger batches strengthen the expected descent toward the fixed point.

3.8 Almost Sure Convergence

From this point onward, the expectation identities from the previous sections are used conditionally on the past filtration $\mathcal{F}_{n-1}$. This is justified because the current iterate q_{n-1} is $\mathcal{F}_{n-1}$ -measurable, while the batch variables E_n and C_n represent fresh sampling randomness at iteration n . Under Assumptions 2 and 3 in their conditional form, Theorem 1 and Theorem 3 remain valid with conditional expectations, which is exactly the form needed for the Lyapunov recursion below.

To apply a stochastic approximation theorem, the second moment of the search direction must also be controlled. Recall that

$$s_n = q_{n-1} - T_{E_{n-1}, C_{n-1}}[q_{n-1}]. \quad (36)$$

Only the coordinates with $|C_{n-1}(s, a)| > 0$ contribute to the update.

Using $(u - v)^2 \leq 2u^2 + 2v^2$, the reward bound $|r| \leq \bar{R}$, and $\max_{a'} q(s', a') \leq \|q\|_\infty$, one obtains

$$\|s_n\|_2^2 \leq 2\|q_{n-1}\|_2^2 + 4(\bar{R}^2 + \gamma\|q_{n-1}\|_\infty^2) \sum_{(s,a) \in \mathcal{S} \times \mathcal{A}} \mathbf{1}_{\{|C_{n-1}(s,a)| > 0\}}. \quad (37)$$

Taking conditional expectations and using Assumption 2 gives

$$\mathbb{E}[\|s_n\|_2^2 \mid \mathcal{F}_{n-1}] \leq 4m\bar{R}^2 + 2(2m\gamma + 1)\|q_{n-1}\|_2^2, \quad (38)$$

where $\mathcal{F}_{n-1}$ denotes the natural filtration and $\|q\|_\infty \leq \|q\|_2$ was used.

Next,

$$\|q_{n-1}\|_2^2 = \|(q_{n-1} - q^*) + q^*\|_2^2 \leq 2\|q_{n-1} - q^*\|_2^2 + 2\|q^*\|_2^2. \quad (39)$$

Since $J(q) = \frac{1}{2}\|q - q^*\|_2^2$, this implies

$$\mathbb{E}[\|s_n\|_2^2 \mid \mathcal{F}_{n-1}] \leq \lambda + K_1 J(q_{n-1}), \quad (40)$$

with admissible constants

$$\lambda = 4m\bar{R}^2 + 4(2m\gamma + 1)\|q^*\|_2^2, \quad K_1 = 8(2m\gamma + 1). \quad (41)$$

Thus, the search direction satisfies the quadratic-growth condition required in standard stochastic approximation results [8, 12].

We now combine the previous ingredients.

Theorem 4. Under Assumptions 1 to 4 and condition (30), the iteration (16) converges almost surely to the optimal fixed point q^* :

$$q_n \rightarrow q^* \quad \text{almost surely as } n \rightarrow \infty. \quad (42)$$

Justification. By Theorem 3, the mean search direction satisfies the pseudogradient estimate (33). By (40), its second moment satisfies the required growth bound. Together with the step-size assumptions (11), these properties verify the hypotheses of a standard stochastic approximation convergence result [7, 12]. Therefore, the sequence (q_n) converges almost surely to q^* . $\square$

Theorem 4 provides the central theoretical guarantee of the report. It shows that the batch-based method also converges to the optimal action-value function under clearly stated assumptions.

3.9 Convergence Rate and the Role of the Batch Size

Beyond convergence itself, the main question of this report is how the batch size affects the speed of convergence. To make this explicit, we use that the quadratic error functional J has 1-Lipschitz gradient and apply the corresponding quadratic upper bound along the iteration (18). This yields the recursion

$$\mathbb{E}[J(q_n) \mid \mathcal{F}_{n-1}] \leq J(q_{n-1}) - \alpha_n \frac{m}{d} (1 - \sqrt{d}\gamma) \|q_{n-1} - q^*\|_2^2 + \frac{1}{2} \alpha_n^2 \mathbb{E}[\|s_n\|_2^2 \mid \mathcal{F}_{n-1}]. \quad (43)$$

Using (19) and (40) yields

$$\mathbb{E}[J(q_n) \mid \mathcal{F}_{n-1}] \leq \left(1 - K_3 \alpha_n + \frac{K_1}{2} \alpha_n^2\right) J(q_{n-1}) + \frac{\lambda}{2} \alpha_n^2, \quad (44)$$

where

$$K_3 = \frac{2m}{d} (1 - \sqrt{d}\gamma). \quad (45)$$

Equation (44) is the key rate relation. The negative drift term, which drives the iteration toward q^* , is proportional to K_3 , and hence proportional to the batch size m . For fixed γ and fixed state-action dimension d , increasing the batch size therefore produces a stronger expected descent per iteration.

This theoretical conclusion is exactly the statement to be tested experimentally. If the analysis captures the essential mechanism of batch sampling, then larger batches should reduce the distance to q^* more quickly than smaller batches.

3.10 Interpretation of the Theoretical Result

The convergence proof establishes three points that are central for the rest of the report. First, the expected empirical Bellman operator is a convex combination of the exact Bellman operator and the identity, as shown in (12). Second, under the condition $\sqrt{d}\gamma < 1$, the

Bellman part of the update forms an acute angle with the error vector, which yields a strictly positive descent toward the fixed point. Third, the strength of this descent is proportional to the batch size m . This restriction on the discount factor is an important limitation of the theorem. In the FrozenLake setting used later in the experiments, $d = 64$, so the condition becomes $\gamma < 1/\sqrt{64} = 1/8 = 0.125$. Consequently, the formal convergence guarantee applies only to the smaller tested discount factors, while the results for larger values of γ must be interpreted as empirical evidence beyond the proven regime.

Furthermore, note that instead of drawing updates from a trajectory-dependent replay buffer, the proof assumes a controlled sampling scheme over state-action coordinates. In practical applications of replay-based Q-learning, however, the batch is generated from the distribution of previously visited transitions [5, 6, 11]. The theory therefore does not model every detail of replay learning, but it does isolate the mechanism of primary interest here, namely that averaging several Bellman targets in a single update strengthens the systematic movement toward q^* .

The experimental section examines this claim through batch-size experiments. In particular, it studies whether larger values of m lead to a faster reduction of the distance to Q^* .

4 Experimental Evaluation

4.1 Experimental Design

The empirical objective of this report is to test the theoretical prediction from Section 3, namely that larger batch sizes should lead to improved convergence behavior. To do so, the experiments use the same model as in the convergence analysis. In each iteration, exactly m distinct state-action pairs are selected uniformly from the full state-action space, and only these coordinates are updated.

A stationary reference solution Q^* is computed by dynamic programming. This reference is not itself part of the experimental comparison. Rather, it provides the target against which the learned action-value estimates are evaluated through the distance $\|Q_t - Q^*\|_2$.

All experiments are carried out on the tabular FrozenLake environment with the 4×4 map and stochastic transitions. The state-action space therefore has dimension

$$d = |\mathcal{S}| |\mathcal{A}| = 16 \cdot 4 = 64. \quad (46)$$

For every tested value of the discount factor γ , a stationary dynamic programming solution Q^* is computed and used as the reference target. The error metric is the Euclidean distance

$$\|Q_t - Q^*\|_2. \quad (47)$$

In the present experiments, improved convergence behavior is assessed through a faster decrease of the distance to the reference solution Q^* . Unless stated otherwise, the reported curves are averaged over 10 random seeds. In the batch-size experiments, shaded regions indicate one standard deviation across these 10 seeds, whereas the cross-gamma comparison shows only mean curves in order to keep the figure readable.

4.2 Testing Batch Sizes

The cleanest empirical test of the theory is obtained by varying the batch parameter m directly in the update rule. To make the effect of the batch size particularly visible in the initial phase of learning, the following experiment focuses on the first 1000 iterations and

uses a common global step-size schedule across all tested values of m . The considered batch sizes are

$$m \in \{1, 4, 8, 16, 32\}, \quad (48)$$

and one sampled Bellman target is used per selected coordinate. Two discount factors are considered: $\gamma = 0.1$ and $\gamma = 0.9$. The first value lies inside the proven regime, since

$$\sqrt{d}\gamma = 8 \cdot 0.1 = 0.8 < 1, \quad (49)$$

whereas the second lies outside it:

$$\sqrt{d}\gamma = 8 \cdot 0.9 = 7.2 > 1. \quad (50)$$

Figures 2 and 3 report the raw distance to Q^* over the first 1000 iterations for the tested values of m . Each panel corresponds to one fixed batch size and shows the mean across multiple random seeds together with one standard deviation.

The most important observation is the behavior in the first updates. For both $\gamma = 0.1$ and $\gamma = 0.9$, the distance to Q^* decreases more rapidly for larger batch sizes than for the non-batched baseline $m = 1$. This effect is especially large for $m = 16$ and $m = 32$, which approach the reference solution more quickly during the initial phase of learning. The short-horizon plots are therefore consistent with the theoretical picture from Section 3, which claims that larger values of m strengthen the expected movement toward Q^* , so the iterate approaches the target more directly in the initial phase.

At the same time, the interpretation should remain careful. The comparison is made *per iteration*, not under a matched computational budget. One iteration with $m = 32$ updates many more coordinates than one iteration with $m = 1$, so the observed advantage should be understood as faster convergence per update step. Moreover, the results for $\gamma = 0.9$ lie outside the formal assumption $\sqrt{d}\gamma < 1$. Even there, however, the same qualitative ordering remains visible, which suggests that the proven condition is sufficient for the convergence argument, but not necessary for useful empirical behavior.

Since the scale of the optimal action-value function depends on γ , the raw distances in Figures 2 and 3 should be interpreted within each fixed discount factor. A normalized comparison across discount factors is therefore given separately in the next subsection.

4.3 Comparison Across Discount Factors

The results above suggest that $m = 32$ provides the strongest empirical improvement among the tested batch sizes. To examine this best-performing batched configuration more systematically across discount factors, Figure 4 compares the normalized distance $\|Q_t - Q^*\|_2 / \|Q^*\|_2$ for the batched update with $m = 32$ and the non-batched baseline with $m = 1$ across the tested discount factors $\gamma \in \{0.1, 0.3, 0.6, 0.9\}$. The curves are means over multiple random seeds. To keep the figure readable, standard-deviation bands are omitted in this cross-gamma comparison.

For every tested value of γ , the curve for $m = 32$ remains below the corresponding baseline curve for $m = 1$ in the initial learning phase, indicating faster convergence toward the reference solution. Thus, the empirical advantage of batch sampling is not confined to a single discount factor, but persists across the tested range of values.

At the same time, the normalized error decreases more slowly for larger discount factors. Taken together, the normalized comparison supports the conclusion that increasing the batch size improves convergence behavior, while larger values of γ make the approximation problem more challenging.

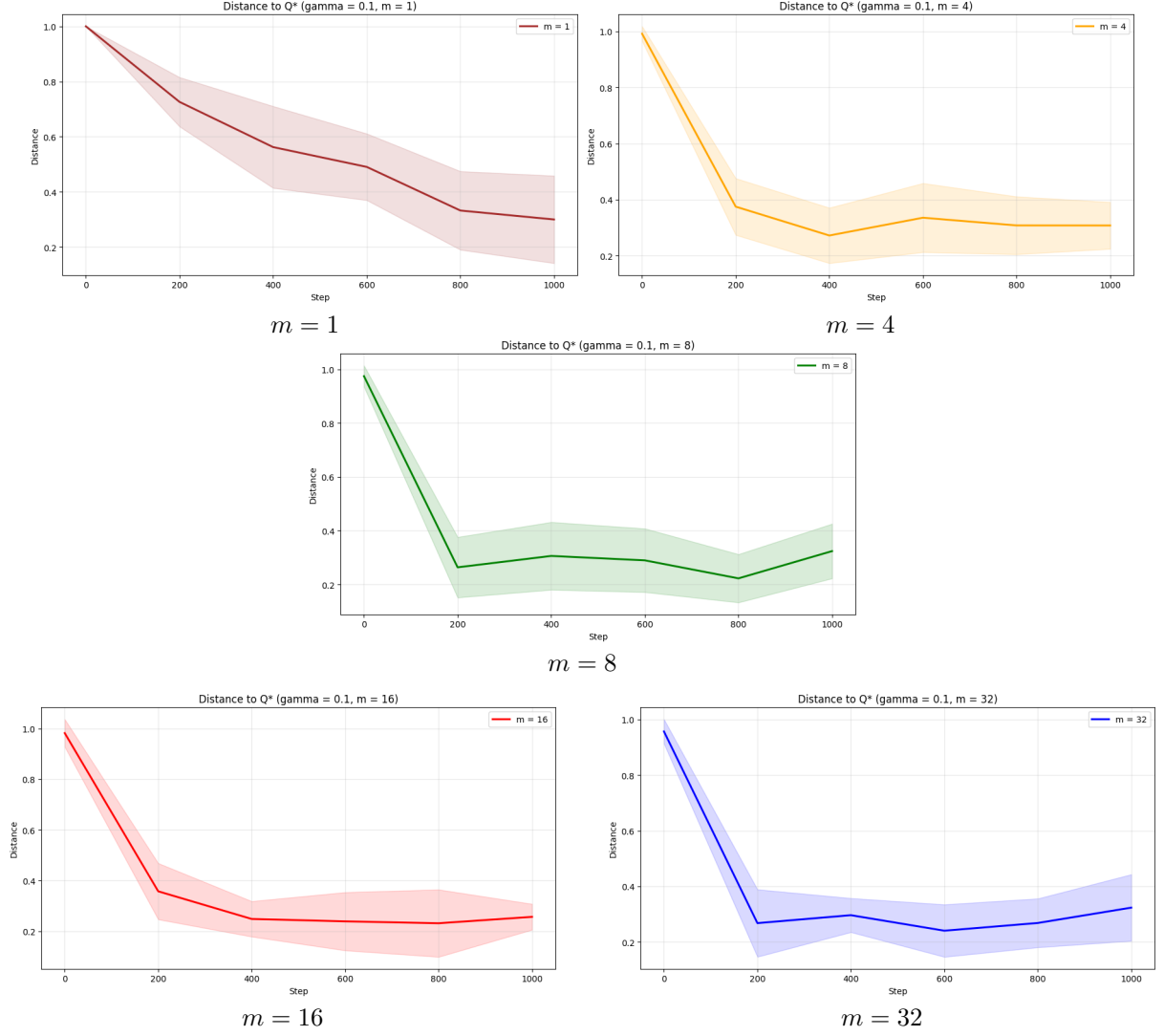


Figure 2: Distance to Q^* during the first 1000 iterations for $\gamma = 0.1$ and $m \in \{1, 4, 8, 16, 32\}$. Each panel shows one batch size. The curves denote multi-seed means, and the shaded regions indicate one standard deviation. Larger batch sizes lead to a faster early decrease of the error.

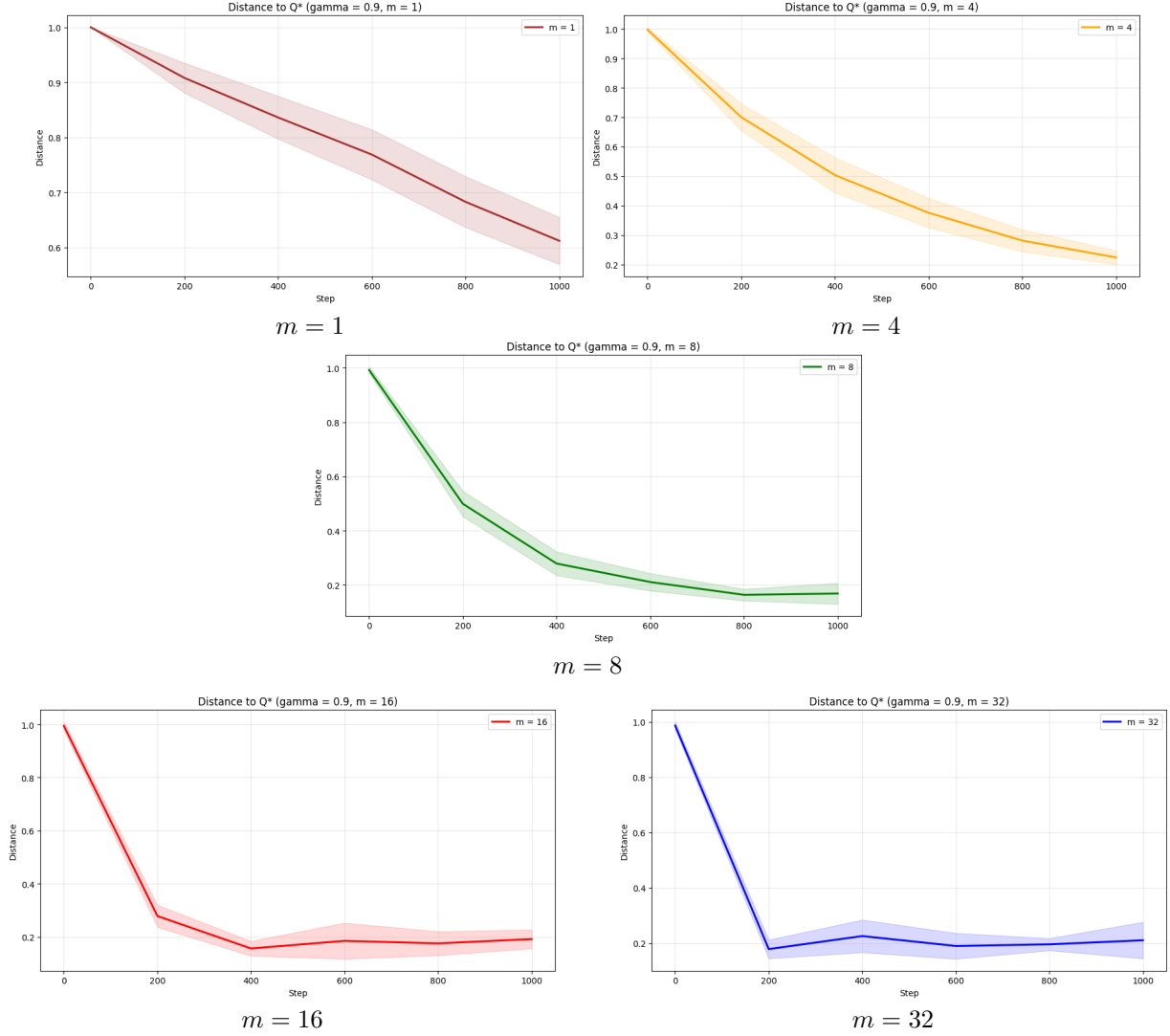


Figure 3: Distance to Q^* during the first 1000 iterations for $\gamma = 0.9$ and $m \in \{1, 4, 8, 16, 32\}$. Each panel shows one batch size. The curves denote multi-seed means, and the shaded regions indicate one standard deviation. The same qualitative ordering remains visible outside the formally proven regime.

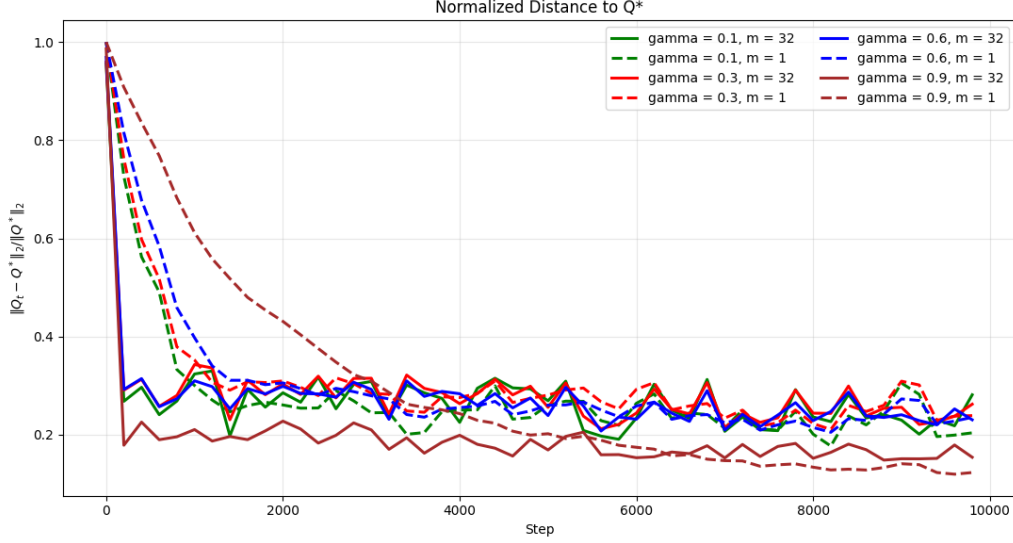


Figure 4: Normalized distance for the batched update with $m = 32$ and the non-batched baseline with $m = 1$ across the tested discount factors $\gamma \in \{0.1, 0.3, 0.6, 0.9\}$. Solid lines correspond to $m = 32$, dashed lines to $m = 1$. All curves show multi-seed means. For every tested value of γ , the batched method approaches the reference solution faster than the non-batched baseline.

5 Discussion

The central question of this report was whether taking multiple samples in a Q-learning update improves convergence behavior. The combined theoretical and empirical results support an answer in favor of the batch-sampling setting studied here.

From the theoretical point of view, the key mechanism is already visible in the expectation identity for the empirical Bellman operator. In expectation, the batch operator is a convex combination of the Bellman optimality operator and the identity, with weight m/d on the Bellman part. This is the point at which the batch size enters the analysis. A larger value of m increases the contribution of the Bellman operator in each iteration and therefore strengthens the movement toward the fixed point q^* .

This effect becomes more explicit in the pseudogradient argument. Under the condition $\sqrt{d}\gamma < 1$, the Bellman image $B[q]$ is sufficiently closer to q^* in the Euclidean norm than the current iterate q . As a consequence, the mean search direction forms a positive angle with the error direction $q - q^*$, and the resulting descent term is proportional to m/d . The same dependence appears again later through the drift constant

$$K_3 = \frac{2m}{d}(1 - \sqrt{d}\gamma). \quad (51)$$

In this sense, the theory also explains why larger batch sizes should accelerate convergence.

The experiments are consistent with this prediction. In the multi-seed batch-size runs, larger values of m lead to a faster decrease of the raw distance to the reference solution Q^* for both $\gamma = 0.1$ and $\gamma = 0.9$. In particular, the larger tested batch sizes, especially $m = 16$ and $m = 32$, outperform the non-batched baseline $m = 1$ in the early phase of learning. The normalized cross-gamma comparison then shows that the advantage of the batched configuration $m = 32$ over the baseline $m = 1$ persists across the tested discount factors. Thus, the empirical results support the central theoretical claim that increasing the batch size improves convergence behavior.

At the same time, the experiments also clarify how this conclusion should be interpreted. The comparison is carried out *per iteration*, not under a matched computational budget. One iteration with $m = 32$ updates many more coordinates than one iteration with $m = 1$, so the observed acceleration should be understood as faster convergence per update step. This is the natural notion of speed for the stochastic-approximation analysis, but it is not the same as a comparison under equal arithmetic effort. A compute-matched analysis would therefore be a natural extension of the present work.

A further point concerns the role of the discount factor. The proof requires the condition $\sqrt{d}\gamma < 1$, which in the FrozenLake setting with $d = 64$ reduces to $\gamma < 1/8$. The theorem therefore applies only to the smaller tested discount factors. Nevertheless, the experiments still show favorable behavior for substantially larger values of γ . Even for $\gamma = 0.9$, larger batch sizes continue to outperform smaller ones. This suggests that the theoretical restriction is conservative in practice.

The normalized cross-gamma plot is particularly important in this regard. It shows that the observed improvement of the batched configuration $m = 32$ over the non-batched baseline $m = 1$ is not merely a consequence of the larger scale of Q^* at higher discount factors. Even after dividing by $\|Q^*\|_2$, the batched method remains below the corresponding baseline across the tested values of γ . This strengthens the interpretation that the batch effect is not an artifact of comparing raw value scales across different discount factors.

At the same time, the experiments suggest a further question about the interaction between the batch size and the discount factor. In the reported plots, the separation between smaller and larger batch sizes appears more pronounced for $\gamma = 0.9$ than for $\gamma = 0.1$, especially in the initial phase of learning. One possible interpretation is that the empirical benefit of larger batches becomes more visible when the approximation problem is more difficult, as is the case for larger discount factors. However, the present experiments are not sufficient to establish a general relationship of this kind. A systematic investigation of how the optimal or most effective batch size depends on γ would therefore be an interesting direction for future work.

Overall, the theoretical and empirical findings tell a coherent story. The stochastic-approximation analysis predicts that larger batch sizes strengthen the descent toward q^* , and the experiments show the corresponding improvement in the observed distance curves. At the same time, the study also reveals the limits of the current analysis. The theoretical condition on γ is conservative, and the notion of speed used in the experiments is per-iteration rather than cost-matched. These limitations do not weaken the main conclusion, but they do indicate promising directions for future work, especially sharper convergence guarantees and cost-sensitive comparisons of different batch sizes.

6 Conclusion

This report studied batch sampling in tabular Q-learning from both a theoretical and an empirical perspective. The central idea was to replace the single stochastic Bellman target of classical Q-learning by an averaged target computed from multiple samples. The aim was to determine whether this averaging improves convergence behavior.

The theoretical analysis was formulated as a stochastic approximation problem for an empirical Bellman operator. It was shown that the expected operator is a convex combination of the Bellman optimality operator and the identity. Using a Euclidean acute-angle argument, a pseudogradient condition and a second-moment bound were established. Under standard boundedness and step-size assumptions, together with the condition $\sqrt{d}\gamma < 1$, almost sure convergence to the optimal action-value function follows. Moreover, the resulting drift constant depends linearly on the batch size m , which explains why larger batches should

accelerate convergence.

The empirical evaluation supported this prediction. In the multi-seed batch-size experiments on FrozenLake, larger values of m led to a faster reduction of the raw distance to Q^* per iteration for both $\gamma = 0.1$ and $\gamma = 0.9$. In particular, the larger tested batch sizes, especially $m = 16$ and $m = 32$, outperformed the non-batched baseline $m = 1$ in the early phase of learning. A normalized cross-gamma comparison for $m = 32$ and $m = 1$ further showed that this qualitative advantage remains visible across the tested discount factors. The experiments also showed that favorable behavior persists even for discount factors outside the formally proven regime, which suggests that the condition $\sqrt{d}\gamma < 1$ is conservative in practice.

The main conclusion of this report is therefore that, in the tabular setting considered here, batch sampling improves the convergence behavior of Q-learning. The theory explains why this effect should occur, and the experiments confirm it in the setting for which the analysis was developed.

Several extensions suggest themselves naturally. A compute-matched comparison could clarify how the improvement per iteration translates into improvement per unit of computational effort. It would also be of interest to derive sharper conditions on the discount factor and to investigate whether similar arguments can be developed for related algorithms such as Speedy Q-learning or for more general function-approximation settings.

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A Appendix

A.1 Hyperparameter Choices

All experiments were carried out on the tabular FrozenLake environment with the 4×4 map and stochastic transitions, that is, with `is_slippery=True`. The state-action space therefore has dimension $d = 64$. For each tested discount factor γ , a dynamic-programming reference solution Q^* was computed and used to evaluate the error $\|Q_t - Q^*\|_2$.

The tested batch sizes were

$$m \in \{1, 4, 8, 16, 32\}.$$

In the short-horizon batch-size experiments, the displayed curves cover the first 1000 iterations in order to highlight the initial convergence behavior for different values of m . The update followed a common global decaying step-size schedule of the form

$$\alpha_t = \frac{\alpha}{1 + t/\alpha_\tau},$$

with initial learning-rate parameter $\alpha = 0.1$ and decay parameter $\alpha_\tau = 20\,000$.

For the normalized comparisons, the plotted quantity was

$$\frac{\|Q_t - Q^*\|_2}{\|Q^*\|_2}.$$

This normalization was used to make results more comparable across discount factors, since the scale of Q^* depends on γ .